American Computer Science League

2023-2024 ● Contest 2: Short Problems ● Classroom Division

1. Prefix/Infix/Postfix Notation

Evaluate the following prefix expression if all numbers are single digits:

− + 2 * 3 8 / * 6 2 3

A. 12
B. 20
C. 22
D. 25
E. 30

2. Prefix/Infix/Postfix Notation

Convert the following infix expression to postfix:

$\frac{A+B}{C-1} + \frac{2A-3}{D}$

A. AB+C1−/2A3*−D/+
B. AB+C1−/2A*3−D/+
C. ABC1−+/2A*3−D/+
D. AB+C1−/2A*3−D+/
E. AB+C1−/2A*3D−/+

3. Bit-String Flicking

Evaluate:

00110 OR 01100 XOR 11010

A. 01001
B. 11001
C. 00110
D. 01010
E. 10110

4. Bit-String Flicking

Evaluate:

(LSHIFT-1 01110) AND (RCIRC-2 (10110 OR NOT 10101))

A. 10100
B. 11111
C. 11100
D. 00011
E. 01001

5. LISP

Evaluate the following sequence of LISP expressions:

(SETQ X '(1 2 (a (b) c) ((1 a) (2 b) c)))
(CDADDR X)

A. (a (b) c)
B. (b)
C. ((b) c)
D. (a (b))
E. (a)

6. Prefix/Infix/Postfix Notation

Evaluate the following postfix expression if all numbers are single digits:

5 8 − 6 9 6 − * 3 / +

A. 6
B. 11
C. 3
D. -2
E. 18

7. Prefix/Infix/Postfix Notation

Translate the following infix expression into prefix:

$$\frac{(a+b)^2-c}{d+e(b^2-a)}$$

A. /−^++ab2cd*e−^b2a
B. /−^ab2c++d*e−^b2a
C. /^−+ab2c+d*e−^b2a
D. /−^+ab2c−d*e+^b2a
E. /−^+ab2c+d*e−^b2a

8. Bit-String Flicking

Evaluate:

(LCIRC-2 (RSHIFT-1 (NOT 10100))) OR (RCIRC-1 (LSHIFT-2 10111))

A. 01011
B. 00101
C. 01111
D. 11110
E. 11111

9. Bit-String Flicking

Solve the following expression for X if X is a 5-bit string:

(LCIRC-2 01101) AND X OR (RSHIFT-1 01111) = 10111

A. 0*1*0
B. 1****
C. 0***1
D. 01*01
E. 00101

10. LISP

Evaluate the following LISP expression:

(CAR (CDR (CAR (CDR '(5 (2 (3 4) 1) (6 7 8) 9)))))

A. (7 8)
B. (6 7 8)
C. (3 4)
D. ((3 4) 1)
E. (2 (3 4))